TABLE 3-4
CONSTRUCTION CLASSIFICATIONS

CONSTRUCTION GROUP I
NONCOMBUSTIBLE

Required fire-resistance ratings of construction elements in hours, based on the test procedures of reference standard RS 3-1.

Key:

N.P. – Not permitted

N.L. – No limit

Noncombustible materials

TABLE 3-4

CONSTRUCTION ELEMENT			CLASS I-A		CLASS I-B		CLASS I-C		CLASS I-D		CLASS I-E	
			Rating in Hrs.	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*
Exterior Walls with an Exterior Separation of	3'-0" or less	Bearing	4	N.P.	3	N.P.	2	N.P.	2	N.P.	2	N.P.
		Non-bearing[f]	2		2		2		2		2	
	More than 3'-0" but less than 15'-0"	Bearing	4	3⅓% protected	3	3⅓% protected	2	3⅓% protected	2	3⅓% protected	2	6⅔% protected
		Non-bearing[f]	2		2		2		2		2	
	15'-0" or more but less than 30'-0"	Bearing	4	3⅓%	3	3⅓%	2	3⅓%	1	3⅓%	0	N.L.
		Non-bearing[f]	1½		1½		1		1		0	
	30'-0" or more	Bearing	4	N.L.	3	N.L.	2	N.L.	1	N.L.	0	
		Non-bearing[f]	0		0		0		0		0	
Interior bearing walls and bearing partitions.			4		3		2		1		0[g,i]	
Enclosure of vertical exits[e], exit passageways, hoistways and shafts.			2		2		2		2		2	
Fire divisions and fire separations.			See Article 5									
Columns, girders, trusses (other than roof trusses) and framing.[k]	Supporting one floor		3		2		1½		1		0[g,i]	
	Supporting more than one floor[l]		4		3		2		1		0[g,i]	
Structural members supporting a wall.			Same as required fire resistance of wall supported, but not less than rating required for member by the class of construction.									
Floor construction including beams.			3		2		1½		1		0[g,i]	
Roof construction including beams, trusses and framing, including arches, domes, shells, cable supported roofs and roof decks[h].	15'-0" or less in ht. above floor to lowest member		2		1½		1[i]		1[i]		0[g,i]	
	15'-0" to 20'-0" in ht. above floor to lowest member		2[c,i] or 1[d,i]		1½[c,i] or 1[d,i]		1[i]		1[i]		0[g,i]	
	20'-0" or more in ht. above floor to lowest member		2[c,i] or 0[d,g,i]		1½[c,i] or 0[d,g,i]		1[c,i] or 0[d,g,i]		1[c,i] or 0[d,g,i]		0[g,i]	

[l] Members supporting loads of not more than two floors or one floor and a roof need not have a fire-resistance rating greater than the floor construction fire-resistance requirement in buildings classified in occupancy groups G, H, and J-2, not including unsprinklered spaces of other occupancies, and in fully sprinklered buildings in occupancy groups E and J-1.

TABLE 3-4
CONSTRUCTION CLASSIFICATIONS
(Continued)

CONSTRUCTION GROUP II
COMBUSTIBLE

Required fire-resistance ratings of construction elements in hours, based on the test procedures of reference standard RS 3-1.

Key:

N.P. – Not permitted

N.L. – No limit

Noncombustible materials

TABLE 3-4 (Continued)

CONSTRUCTION ELEMENT			CLASS II-A		CLASS II-B		CLASS II-C		CLASS II-D		CLASS II-E	
			Rating in Hrs.	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*	*Rating in Hrs.*	*Ext.[a,b] Open'g*
Exterior Walls with an Exterior Separation of	3'-0" or less	Bearing	2	N.P.	2	N.P.	2	N.P.	2	N.P.	2	N.P.
		Non-bearing[f]	2		2		2		2		2	
	More than 3'-0" but less than 15'-0"	Bearing	2	3⅓ % protected	2	3⅓ % protected	2	3⅓ % protected	1	6⅔ %	1	6⅔ %
		Non-bearing[f]	2		2		2		1		1	
	15'-0" or more but less than 30'-0"	Bearing	2	3⅓ %	2	3⅓ %	2	3⅓ %	1	N.L.	0	N.L.
		Non-bearing[f]	2		2		2		1		0	
	30'-0" or more	Bearing	1	N.L.	1½	N.L.	1½	N.L.	1		0	
		Non-bearing[f]	0		0		0		0		0	
Interior bearing walls and bearing partitions.			2		1		0		1		0	
Enclosure of vertical exits[e], exit passageways, hoistways and shafts.			2		2		1[i]		1[i]		1	
Fire divisions and fire separations.			See Article 5									
Columns, girders, trusses (other than roof trusses) and framing.[k]	Supporting one floor		see section 27-623		1		0 or 1[j]		1		0	
	Supporting more than one floor		see section 27-623		1		0 or 1[j]		1		0	
Structural members supporting a wall.			Same as required fire resistance of wall supported, but not less than rating required for member by the class of construction.									
Floor construction including beams.			see section 27-623		1		0 or 1[j]		1		0	
Roof construction including beams, trusses and framing, including arches, domes, shells, cable supported roofs and roof decks[h].	15'-0" or less in ht. above floor to lowest member		see section 27-623		¾		0		¾		0	
	15'-0" to 20'-0" in ht. above floor to lowest member		see section 27-623		¾		0		¾		0	
	20'-0" or more in ht. above floor to lowest member		see section 27-623		¾		0		¾		0	